

Database Technologies-Assignment 8

Use sub-query to solve following problems.

Note: To solve below queries, use “sales” database

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
KD2_98846_Raj>SELECT * FROM orders WHERE cnum = (SELECT cnum FROM customers
WHERE cname= 'Cisneros');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+-----+-----+-----+-----+-----+
2 rows in set (0.04 sec)
```

7. Extract all the orders of Motika.

```
KD2_98846_Raj>SELECT * FROM orders WHERE snum = (SELECT snum FROM salespeople WHERE sname = 'Motika');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
KD2_98846_Raj>SELECT * FROM orders WHERE snum IN(SELECT snum FROM customers WHERE city = 'London' );
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
3 rows in set (0.01 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
KD2_98846_Raj>SELECT * FROM orders WHERE amt>ANY(SELECT amt FROM orders where cnum in(select cnum from customers WHERE city = 'London'));
```

onum	amt	odate	cnum	snum
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3011	9891.88	1990-10-04	2006	1001

6 rows in set (0.01 sec)

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'.
(Hint: find all customers of serres salesperson).

```
KD2_98846_Raj>SELECT * FROM customers WHERE rating>=ANY(SELECT rating FROM customers WHERE snum IN(SELECT snum FROM salespeople WHERE sname = 'Serres'));
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.01 sec)

11. Select customers who have a greater rating than any other customer in Rome.

```

KD2_98846_Raj>SELECT *
-> FROM customers
-> WHERE rating > ANY (
->     SELECT rating
->     FROM customers
->     WHERE city = 'Rome'
-> );

```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.00 sec)

12. Select all orders that had amounts that were greater than at least one of the orders from '1990-10-04'.

```

KD2_98846_Raj>SELECT *
-> FROM orders
-> WHERE amt > ANY (
->     SELECT amt
->     FROM orders
->     WHERE odate = '1990-10-04'
-> );

```

onum	amt	odate	cnum	snum
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002
3011	9891.88	1990-10-04	2006	1001

8 rows in set (0.01 sec)

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```

KD2_98846_Raj>SELECT *
-> FROM orders
-> WHERE amt < ANY (
->     SELECT o.amt
->     FROM orders o
->     JOIN customers c
->     ON o.cnum = c.cnum
->     WHERE c.city = 'San Jose'
-> );

```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3007	75.75	1990-10-04	2004	1002
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002

8 rows in set (0.01 sec)

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
KD2_98846_Raj>SELECT o.amt, c.cname, c.rating
-> FROM orders o
-> JOIN customers c
-> ON o.cnum = c.cnum
-> WHERE o.amt > (
->     SELECT AVG(amt)
->     FROM orders
-> );
```

amt	cname	rating
5160.45	Liu	200
9891.88	Clemens	100
4723.00	Clemens	100

3 rows in set (0.00 sec)

14. Select those customers whose rating are higher than every customer in London.

```
KD2_98846_Raj>SELECT o.amt, c.cname, c.rating
-> FROM orders o
-> JOIN customers c
-> ON o.cnum = c.cnum
-> WHERE o.amt > (
->     SELECT AVG(amt)
->     FROM orders
-> );
```

amt	cname	rating
5160.45	Liu	200
9891.88	Clemens	100
4723.00	Clemens	100

3 rows in set (0.00 sec)

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
KD2_98846_Raj>SELECT s.sname, s.city
-> FROM salespeople s
-> WHERE s.city <> ALL (
->     SELECT c.city
->     FROM customers c
-> );
```

sname	city
Rifkin	Barcelona
Axelrod	New York

2 rows in set (0.03 sec)

9. Find names and numbers of all salesperson who have more than one customer.

```
KD2_98846_Raj>SELECT snum, sname
-> FROM salespeople
-> WHERE snum IN (
->     SELECT snum
->     FROM customers
->     GROUP BY snum
->     HAVING COUNT(*) > 1
-> );
```

snum	sname
1001	Peel
1002	Serres

2 rows in set (0.04 sec)

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```

KD2_98846_Raj>SELECT s.snum, s.sname, s.city,
->      (
->          SELECT COUNT(*)
->          FROM customers c
->          WHERE c.snum = s.snum
->      ) AS customer_count
-> FROM salespeople s
-> WHERE s.snum IN (
->     SELECT snum
->     FROM customers
->     GROUP BY snum
->     HAVING COUNT(*) > 1
-> );

```

snum	sname	city	customer_count
1001	Peel	London	2
1002	Serres	San Jose	2

2 rows in set (0.01 sec)

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```

KD2_98846_Raj>SELECT snum, SUM(amt) AS total_amount
-> FROM orders
-> GROUP BY snum
-> HAVING SUM(amt) > (
->     SELECT MAX(amt)
->     FROM orders
-> );

```

snum	total_amount
1001	15382.07

1 row in set (0.04 sec)

